

BEN ZHOU

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EDUCATION

University of Waterloo

Candidate for BAsC in Computer Engineering, Honours

Waterloo, Ontario

Sep 2024 – Apr 2029

EXPERIENCE

Grand Charter

Founding Engineer

New York City, New York

May 2026 – Present

- Led a redesign of **3 core apps**, cleaning code and standardizing UI systems across **2,500 files** and **250,000 LOC**.
- Provided **FDE** services for many **enterprise clients**, shipping solutions and directly augmenting their systems.
- Reduced exec/ops overview load times from **35s** → **0.6s** by profiling queries, and eliminating inefficient query patterns.
- Improved telemetry and lead tracking data by accurately determining sources, and consolidating/backfilling older cases.
- Deployed custom internal tools for data analysis over **100k** rows and general hiring workflows.

Software Engineering Intern

Sep 2025 – Apr 2026

- Built integrations for Gmail & Outlook, replicating core app features and cutting document ingestion time by over **78%**.
- Created landing page templates and new **CMS environments**, increasing impressions and clicks by **20–30%** monthly.
- Built dev tools: CI linters, Sanity migration utils, and Anvil form generation, reducing repetitive workflows.
- Built **end-to-end intake pipelines** for **multiple case types** with native integrations and internal tooling.

Fuego.io

Software Engineering Intern

San Francisco, California

Jan 2025 – Apr 2025

- Optimized core AI generation features by developing custom architecture, accelerating response times to be **17.7x faster** (**15.4s** → **0.87s avg**), reducing token usage and slashing costs by similar margins.
- Built an internal dashboard to log **1,000+** AI threads with tagging and analytics, cutting debugging time by **50%**.
- Built **audience segmentation** features, enabling targeted notifications for brands reaching **millions** of users.
- Automated currency conversions with a **GraphQL** CRON job syncing **50 exchange rates** automatically.
- Built analytics tools processing **100k+** **queries** with filters and shareable URLs, cutting reporting time by **40%**.

SAP

Software Developer Co-op

Toronto, Ontario

Feb 2024 – Jul 2024

- Created a worker to serve **GPT-4o** requests using **TypeScript**, reducing request errors by **23%**.
- Wrote scripts in **Python** and **Node** to clean **248,000** records and updated schema to accelerate processing by **26%**.

PROJECTS

🔗 **Lattice** | [GitHub](#) | [Next.js](#), [TypeScript](#), [Bun](#)

- Built a **hybrid retrieval** engine over long PDFs, fusing vector embeddings and **BM25** lexical scoring via **reciprocal rank fusion (RRF)** with reranking and diversification to surface evidence passages.
- Engineered a section-aware ingestion pipeline (PDF extraction, overlapping chunking, summary/keyword enrichment, embedding) feeding a citation-enforced answer layer grounded with a **1.5B-param** model.

🔗 **CrimeX** | [GitHub](#) | [Next.js](#), [Supabase](#), [TypeScript](#)

- Built a geospatial crime platform visualizing **100k+** **incidents** with real-time map analytics by extracting ARCGIS data.
- Developed a forecasting pipeline generating **thousands of crime hotspot predictions** across **4h** and **12h horizons**, with automated evaluation against new incident data.

🔗 **bTagScript Language and Scraper** | [GitHub](#) | [Project](#) | [TypeScript](#), [Next.js](#), [Flask](#), [Python](#), [Mongo](#), [MySQL](#)

- Wrote an **interpreter-based engine** for my own language for instant script execution.
- Deployed a scraper to monitor over **2 million scripts** and archive an additional **35,900+** user-uploaded scripts.
- Designed a **RESTful API** to run sandboxed programs, logging runtime data through **MySQL** for analytics and debugging.
- Processed **16,000+** **programs**, and recently migrated to a new host to ensure **99.98%** uptime for all users.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, HTML, CSS, C++, C, Python, Java, SQL, GraphQL, Rust

Tech: Next.js, React, Vite, Flask, Express, Mongo, Redis, Tailwind, SQLite, PostgreSQL, Prisma, Drizzle

Development: GCP, AWS (S3, Lambdas), PostHog, Cloudflare, Firebase, Vercel, Vitest, Docker, GitHub Actions, Supabase